

User Acceptance Testing (UAT) Document

Date	30th September
Team ID	SWTID-2026-3342
Project Name	AI FAQ Assistant API

Project Description: The AI FAQ Assistant API is a backend application designed to automate customer support by providing quick answers to frequently asked questions. It allows users to search and retrieve FAQs, access category-based information, and receive AI-generated answers through Google Gemini AI. Administrators and authorized customer care executives can manage FAQ content. The application is developed using Node.js, Express.js, MongoDB, Mongoose, JWT, and bcrypt.

Project Version: Version 1.0

Testing Period: [Start Date] to [End Date]

3. Testing Scope

3.1 Features and Functionalities to Be Tested

- User registration and input validation.
- User login and authentication.
- JWT token generation and validation.
- FAQ creation, retrieval, updating, and deletion.
- FAQ search using keywords.
- FAQ retrieval based on categories.
- AI-powered answer generation using Google Gemini AI.
- API response format and error handling.
- Authorization for protected API endpoints.
- Database operations and data consistency.

3.2 User Stories or Requirements to Be Tested

User Story ID	Requirement
USN-1	User registration
USN-2	User login
USN-3	Access protected endpoints using JWT
USN-4	Search FAQs using keywords
USN-5	View FAQ details and answers

User Story ID	Requirement
USN-6	Search FAQs by category
USN-7	Receive AI-generated answers
USN-8	Receive appropriate error messages
USN-9	Add new FAQs
USN-10	Update existing FAQs
USN-11	Delete FAQs
USN-12	Manage FAQ categories
USN-13	Restrict operations to authorized users
USN-15	Receive consistent API responses

4. Testing Environment

Field	Details
URL / Location	Local development server, e.g. http://localhost:3000
Backend	Node.js and Express.js
Database	MongoDB
Database ODM	Mongoose
API Testing Tool	Postman
Authentication	JWT and bcrypt
External AI Service	Google Gemini API
Credentials	Use valid test user credentials
Test Database	Development or dedicated test database

Note: Replace the example URL and environment details with the actual configuration used for your project. Do not include real passwords, JWT secrets, or API keys in the report.

5. Test Cases

The following test cases cover both positive and negative scenarios for the AI FAQ Assistant API.

Initial test status: Not Run. Record the actual result and mark Pass or Fail after executing each test.

Test Case ID	Test Scenario	Test Steps	Expected Result	Actual Result	Pass/Fail
TC-001	User registration with valid details	1. Open Postman. 2. Send a registration request with valid user details. 3. Check the response.	User account is created successfully, and a success response is returned.	To be recorded during testing.	Not Run
TC-002	Registration with missing fields	1. Send a registration request without a required field. 2. Check the response.	Request is rejected with an appropriate validation error.	To be recorded during testing.	Not Run
TC-003	Login with valid credentials	1. Send a login request with valid credentials. 2. Check the response.	Login succeeds and an authentication token is returned, if supported by the implementation.	To be recorded during testing.	Not Run
TC-004	Login with invalid credentials	1. Send a login request with an incorrect password. 2. Check the response.	Login is rejected with an appropriate error message.	To be recorded during testing.	Not Run
TC-005	Access protected endpoint without token	1. Send a request to a protected endpoint. 2. Do not include a JWT.	Access is denied with an appropriate authentication error.	To be recorded during testing.	Not Run
TC-006	Access protected endpoint with valid token	1. Log in. 2. Copy the valid JWT. 3. Send a request to a protected endpoint with the token.	The request is authorized and processed successfully, subject to permissions.	To be recorded during testing.	Not Run
TC-007	Create a new FAQ	1. Authenticate as an authorized user. 2. Send a request with FAQ question and answer. 3. Check the response.	FAQ is saved in MongoDB and a success response is returned.	To be recorded during testing.	Not Run
TC-008	Retrieve existing FAQs	1. Send a request to retrieve FAQs. 2. Check the returned data.	The API returns the available FAQ records in the expected JSON format.	To be recorded during testing.	Not Run
TC-009	Search FAQ using a valid keyword	1. Send a search request with a keyword present in an FAQ. 2. Check the results.	Matching FAQs are returned.	To be recorded during testing.	Not Run
TC-010	Search FAQ using a non-matching keyword	1. Send a search request with a keyword that does not match any FAQ. 2. Check the response.	The API returns an appropriate empty result or not-found response.	To be recorded during testing.	Not Run

Test Case ID	Test Scenario	Test Steps	Expected Result	Actual Result	Pass/Fail
TC-011	Retrieve FAQs by category	1. Send a request with a valid category. 2. Check the returned FAQs.	FAQs belonging to the requested category are returned.	To be recorded during testing.	Not Run
TC-012	Update an existing FAQ	1. Authenticate as an authorized user. 2. Send an update request for an existing FAQ. 3. Retrieve the FAQ again.	The FAQ is updated, and the new details are returned.	To be recorded during testing.	Not Run
TC-013	Delete an existing FAQ	1. Authenticate as an authorized user. 2. Send a delete request for an existing FAQ. 3. Retrieve the FAQ again.	The FAQ is deleted, and the API confirms the operation.	To be recorded during testing.	Not Run
TC-014	Update a non-existent FAQ	1. Send an update request with an invalid FAQ ID. 2. Check the response.	The API returns an appropriate not-found or validation error.	To be recorded during testing.	Not Run
TC-015	Generate an AI-powered answer	1. Submit a question to the AI answer endpoint. 2. Wait for the Gemini API response. 3. Check the answer.	A generated answer is returned in the expected JSON response.	To be recorded during testing.	Not Run
TC-016	Handle AI service failure	1. Simulate an unavailable AI service or invalid test configuration. 2. Submit a question. 3. Check the response.	The API handles the failure gracefully and returns an appropriate error response.	To be recorded during testing.	Not Run
TC-017	Submit an invalid API request	1. Send a request with malformed or invalid input. 2. Check the response.	The API rejects the request with an appropriate error message and status code.	To be recorded during testing.	Not Run
TC-018	Verify JSON response format	1. Send a valid API request. 2. Inspect the response headers and body.	The response follows the expected JSON structure and uses an appropriate HTTP status code.	To be recorded during testing.	Not Run

6. Bug Tracking

Record any defects identified while executing the test cases. The table below is a blank tracking log; no bugs are assumed to have been found.

Bug ID	Bug Description	Steps to Reproduce	Severity	Status	Additional Feedback
BG-001	To be recorded if a defect is identified.	Record the steps that reproduce the issue.	Low / Medium / High	Open	Add tester observations.
BG-002	To be recorded if a defect is identified.	Record the steps that reproduce the issue.	Low / Medium / High	Open	Add tester observations.

Bug ID	Bug Description	Steps to Reproduce	Severity	Status	Additional Feedback
BG-003	To be recorded if a defect is identified.	Record the steps that reproduce the issue.	Low / Medium / High	Open	Add tester observations.

Bug Severity Guidelines

- High: A major feature is unusable, authentication is compromised, or a critical operation fails.
- Medium: A feature works incorrectly or has a workaround, but the issue affects usability or functionality.
- Low: A minor issue, such as a cosmetic defect or a non-critical message problem.

Bug status can be updated to Open, In Progress, or Closed as the issue is investigated and resolved.

7. Sign-off

The sign-off confirms that the UAT has been completed and the results have been reviewed.

Field	Details
Tester Name	Enter Tester Name
Date of Test Completion	Enter Date
Signature	Tester's Signature
Project Manager Approval	Name and Signature
Product Owner Approval	Name and Signature

8. Notes

- All test cases should be executed in the designated testing environment.
- Both positive and negative test scenarios should be tested.
- Actual results must be recorded after executing each test case.
- Mark a test case as Pass only when the actual result matches the expected result. Otherwise, mark it as Fail and record the defect.
- All identified bugs should be documented with reproduction steps, severity, and status.
- Testers should provide feedback and suggestions for improving the application.
- Obtain sign-off from both the project manager and product owner before proceeding with deployment.

9. Conclusion

User Acceptance Testing for the AI FAQ Assistant API verifies that the application meets its functional requirements and is ready for its intended users. The testing process covers user authentication, FAQ management, keyword and category-based search, AI-powered answer generation, and error handling.

Successful completion of these test cases will provide evidence that the API behaves as expected in the tested scenarios. Any defects identified during testing should be resolved and retested before final approval and deployment.